

## Section 1 - Identification

**Product Name** Magnesium chloride hexahydrate  
**CAS No** 7791-18-6  
**Synonyms** Magnesium dichloride hexahydrate

**Product Code** 197530000; 197530010; 197530050; 197530250  
**Address** ThermoFisher Scientific Australia Pty Ltd  
 5 Caribbean Drive, Scoresby  
 VICTORIA 3179, Australia  
**Emergency Tel.** CHEMTREC®  
 03 9757 4559 or +613 9757 4559  
**Telephone / Fax Numbers** Tel: 1300 735 292  
 Fax: 1800 067 639  
**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.  
 This product does not contain any substance(s) subject to Prohibition, Authorization or  
 Restriction. This product does not contain any substance(s) listed on the voluntary National  
 Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as not hazardous according to criteria of Safe Work Australia.

Physical hazards  
 No hazards identified

Health hazards  
 No hazards identified

Environmental hazards  
 No hazards identified

Label Elements None required

**Other information**  
 No information available

## Section 3 - Composition and Information on Ingredients

| Component                       | CAS No    | Weight % |
|---------------------------------|-----------|----------|
| Magnesium chloride, hexahydrate | 7791-18-6 | >95      |
| Magnesium chloride              | 7786-30-3 | -        |

## Section 4 - First Aid Measures

|                                            |                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Self-Protection of the First Aider</b>  | No special precautions required.                                                                                        |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                    |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                            |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                  |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Chlorine, Magnesium oxides, Hydrogen chloride gas.

### Decomposition Temperature

> 106°C

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### Environmental Precautions

Should not be released into the environment.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**Clean-up methods - large spillage**

Not applicable, packaged goods.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a cool, well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

None under normal use conditions.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Natural rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |                 |                       |
| Neoprene       |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Particle filter (or AUS/NZ equivalent)

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                         |                                          |
|------------------------------------------------|-----------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | White                                   |                                          |
| <b>Physical State</b>                          | Powder Solid                            |                                          |
| <b>Odor</b>                                    | Odorless                                |                                          |
| <b>Odor Threshold</b>                          | No data available                       |                                          |
| <b>pH</b>                                      | 5-6.5                                   | 5% aq. solution                          |
| <b>Melting Point/Range</b>                     | 117 °C / 242.6 °F                       |                                          |
| <b>Softening Point</b>                         | No data available                       |                                          |
| <b>Boiling Point/Range</b>                     | No information available                |                                          |
| <b>Flash Point</b>                             | No information available                |                                          |
| <b>Evaporation Rate</b>                        | Not applicable                          | <b>Method -</b> No information available |
| <b>Flammability (solid,gas)</b>                | No information available                | Solid                                    |
| <b>Explosion Limits</b>                        | No data available                       |                                          |
| <b>Vapor Pressure</b>                          | No data available                       |                                          |
| <b>Vapor Density</b>                           | Not applicable                          | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available                       |                                          |
| <b>Bulk Density</b>                            | No data available                       |                                          |
| <b>Water Solubility</b>                        | 2350 g/L (20°C)                         |                                          |
| <b>Solubility in other solvents</b>            | No information available                |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                         |                                          |
| <b>Autoignition Temperature</b>                | Not applicable                          |                                          |
| <b>Decomposition Temperature</b>               | > 106°C                                 |                                          |
| <b>Viscosity</b>                               | Not applicable                          | Solid                                    |
| <b>Explosive Properties</b>                    | No information available                |                                          |
| <b>Oxidizing Properties</b>                    | No information available                |                                          |
| <b>Other information</b>                       |                                         |                                          |
| <b>Molecular Formula</b>                       | Cl <sub>2</sub> Mg . 6 H <sub>2</sub> O |                                          |
| <b>Molecular Weight</b>                        | 203.31                                  |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                                    |
|-----------------------------------------|----------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available         |
| <b>Stability</b>                        | Stable under normal conditions.                    |
| <b>Conditions to Avoid</b>              | Avoid dust formation.                              |
| <b>Incompatible Materials</b>           | Metals.                                            |
| <b>Hazardous Decomposition Products</b> | Chlorine. Magnesium oxides. Hydrogen chloride gas. |

Hazardous Polymerization

No information available.

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

No data available

Inhalation

No data available

| Component                       | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation |
|---------------------------------|---------------------------|---------------------------|-----------------|
| Magnesium chloride, hexahydrate | LD50 = 8100 mg/kg ( Rat ) |                           |                 |
| Magnesium chloride              | LD50 = 2800 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs

No information available.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available

## Section 12 - Ecological Information

### Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component          | Freshwater Fish                            | Water Flea           | Freshwater Algae    | Microtox                                                                       |
|--------------------|--------------------------------------------|----------------------|---------------------|--------------------------------------------------------------------------------|
| Magnesium chloride | Pimephales promelas:<br>EC50: 2.12 g/L:96H | EC50 : 1400 mg/L/24h | EC50: 2200 mg/L/72h | EC50 Pseudomonas putida: EC50:26,14 g/L/h<br>Photobacterium phosphoreum: EC50: |

|  |  |  |  |                                                                            |
|--|--|--|--|----------------------------------------------------------------------------|
|  |  |  |  | 36,3 mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50:<br>77,2 mg/L/24 h |
|--|--|--|--|----------------------------------------------------------------------------|

**Persistence and Degradability****Persistence**

Soluble in water, Persistence is unlikely, based on information available.

**Degradability**

Not relevant for inorganic substances.

**Bioaccumulative Potential**

Bioaccumulation is unlikely Bioaccumulation is unlikely

**Mobility**

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

**Other Information**

Chemical wastes should be disposed through a licensed commercial waste collection service.

## Section 14 - Transport Information

**IMDG/IMO**

Not regulated

**ADG**

Not regulated

**IATA**

Not regulated

**Environmental hazards**

No hazards identified

**Special Precautions**

No special precautions required

**Additional information**

None known

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National Regulations****Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                                   | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|---------------------------------------------|-------------------------------------------------------------|------------------------|
| Magnesium chloride, hexahydrate - 7791-18-6 | Present                                                     | -                      |
| Magnesium chloride - 7786-30-3              | Present                                                     | -                      |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component                       | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|---------------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Magnesium chloride, hexahydrate | X    | X     | -         | -      | -    | -   | -    | X     | X    | X    | X     | -        |
| Magnesium chloride              | X    | X     | 232-094-6 | -      | X    | X   | -    | X     | X    | X    | X     | KE-22691 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component           | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Magnesium chloride, | 7791-18-6 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

|                    |           |        |                |                |                |
|--------------------|-----------|--------|----------------|----------------|----------------|
| hexahydrate        |           |        |                |                |                |
| Magnesium chloride | 7786-30-3 | Listed | Not applicable | Not applicable | Not applicable |

Authorisation/Restrictions according to EU REACH

Not applicable

## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)            |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  |                                                                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Revision Date** 17-Nov-2022  
**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet